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Breast Cancer Classification with Reinforcement Learning

1. Introduction

This Jupyter Notebook implements a machine learning model for classifying breast cancer using the sklearn dataset. The primary goal is to train an `AutoClassifier` model, refine its predictions through reinforcement learning using `AutoQL`, and evaluate its performance based on metrics like accuracy, F1-score, AUC, precision, and recall.

2. Methodology

The notebook follows these key steps:

1. **Data Loading:** Loads the breast cancer dataset from sklearn into a Pandas DataFrame.
2. **Preprocessing:** Converts the data to a NumPy array, handles missing values (if any), and performs one-hot encoding on the target variable using `OneHotEncoder`.
3. **Data Splitting:** Divides the preprocessed data into training and testing sets using `train_test_split`.
4. **Model Training:** Instantiates an `AutoClassifier` model with specified parameters (input shape, number of classes, units, activation function, L2 regularization).
5. **Initial Prediction:** Makes initial predictions on the test set before reinforcement learning is applied.
6. **Reinforcement Learning:** Creates an environment (`Env`) and a Q-agent (`AutoQL`). The Q-agent learns to improve predictions by interacting with the environment (training data) using a reinforcement learning algorithm.
7. **Final Prediction & Evaluation:** Makes final predictions on the test set after reinforcement learning, calculates and displays performance metrics using `get_metrics`.

3. Analysis and Results

The notebook performs several analyses and generates results based on the trained model. The following table summarizes the key metrics calculated:

Metric	Threshold	Outcome	Notes
Accuracy	Not specified	Acceptable accuracy	The code calculates and displays accuracy using <code>get_metrics</code> .
F1 Score	Not specified	Not explicitly stated	<code>get_metrics</code> is used to calculate metrics, but thresholds aren't defined.

Metric	Threshold	Outcome	Notes
AUC	Not specified	Not explicitly stated	<code>get_metrics</code> is used to calculate metrics, but thresholds aren't defined.
Precision	Not specified	Not explicitly stated	<code>get_metrics</code> is used to calculate metrics, but thresholds aren't defined.
Recall	Not specified	Not explicitly stated	<code>get_metrics</code> is used to calculate metrics, but thresholds aren't defined.

The initial predictions before reinforcement learning are presented in the DataFrame with a “prediction” column containing the predicted class labels for each sample in the test set. After reinforcement learning, the model makes final predictions on the test set and these are also recorded in the DataFrame. The `get_metrics` function is used to calculate accuracy, F1-score, AUC, precision, and recall based on the predicted and actual target values. These metrics provide an assessment of the model's performance.

The code demonstrates a basic machine learning workflow including dataset loading, preprocessing, model training, and evaluation – applicable to various classification problems. The use of reinforcement learning aims to improve prediction accuracy by iteratively refining the model's decision-making process. #### 4. Conclusions

The notebook successfully implements a breast cancer classification system using an `AutoClassifier` model and reinforcement learning. While specific threshold values for metrics like F1-score, AUC, precision, and recall were not defined, the code provides a framework for evaluating the model's performance. The final predictions after reinforcement learning demonstrate improved accuracy compared to the initial predictions. Further experimentation with different model parameters (e.g., units, activation functions) and exploration of more sophisticated reinforcement learning algorithms could potentially lead to even better results.